

ROGERS, AR, USA

WMO: 723449

Lat: 36.372N

Lon: 94.107W

Elev: 1353

StdP: 13.99

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53954

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	15.7	0.1	5.8	13.6	5.0	7.5	20.0	26.2	56.6	23.8	52.3	6.9	10	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	94.7	74.0	91.8	74.4	89.9	73.9	77.7	88.6	76.6	87.2	75.4	85.7	9.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	135.5	84.0	73.2	129.9	82.7	72.3	125.7	81.6	42.1	88.6	40.9	87.2	39.8	85.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.6	19.5	17.5	3.8	97.5	3.8	3.8	1.0	100.2	-1.2	102.4	-3.3	104.5	-6.1	107.2		
			2.2	78.8	4.1	1.5	-0.7	79.9	-3.1	80.8	-5.3	81.6	-8.3	82.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	36.7	40.9	48.5	57.6	65.8	74.7	78.7	77.7	70.8	59.7	48.9	39.3	
	DBStd	17.09	11.63	11.04	10.51	8.58	7.63	5.77	5.24	5.09	7.50	8.68	9.55	10.30	
	HDD50	1417	432	292	151	30	3	0	0	0	1	19	132	357	
	HDD65	3929	876	676	516	247	84	6	0	1	32	207	486	798	
	CDD50	4468	20	36	105	257	494	740	888	858	625	321	99	25	
	CDD65	1506	0	1	5	24	111	295	424	393	206	44	3	0	
	CDH74	13871	0	5	35	152	817	2683	4372	3751	1733	311	12	0	
	CDH80	5505	0	1	2	10	178	1007	1962	1652	643	50	0	0	
Wind	WSAvg	8.0	8.8	9.0	9.4	9.9	8.1	7.4	6.5	6.3	6.6	7.3	8.6	8.3	
Precipitation	PrecAvg	45.4	2.5	2.2	3.6	4.4	5.9	4.7	3.8	2.8	4.3	3.3	4.3	3.1	
	PrecMax	65.5	7.2	5.9	7.8	12.5	11.8	12.3	10.6	6.1	10.4	7.0	9.7	12.8	
	PrecMin	32.4	0.2	0.3	0.8	1.5	1.5	1.1	0.9	0.0	0.5	1.0	0.6	0.3	
	PrecStd	8.1	1.8	1.5	1.7	2.5	2.2	2.8	2.2	1.5	2.3	1.5	2.7	2.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.9	72.1	78.3	81.4	87.8	94.5	99.2	99.1	95.0	84.3	75.2	68.5	
		MCWB	55.6	59.5	62.6	64.6	70.7	74.4	73.0	72.8	71.9	70.0	62.7	59.6	
	2%	DB	63.6	66.4	73.2	78.5	84.2	91.2	95.0	94.3	90.1	81.1	70.5	63.9	
		MCWB	54.8	55.4	58.9	63.2	70.0	74.5	74.6	73.4	72.1	66.7	59.8	55.9	
	5%	DB	59.1	63.0	69.6	75.2	81.7	88.5	92.1	91.2	87.3	77.5	67.2	60.3	
		MCWB	50.9	53.8	57.1	61.9	69.0	73.6	75.3	73.7	71.4	64.2	58.4	53.0	
	10%	DB	54.6	58.8	65.6	72.3	79.0	85.9	89.8	88.8	83.6	73.4	63.9	55.0	
		MCWB	47.4	50.1	55.3	60.5	67.7	73.0	75.0	73.2	69.9	63.1	55.1	48.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	62.2	65.0	68.4	74.2	78.2	79.4	78.9	76.1	72.5	66.3	61.6
			MCDB	65.1	68.8	75.4	77.1	83.2	89.3	90.3	89.6	87.8	81.5	72.3	66.7
2%		WB	57.0	58.9	62.0	66.3	72.3	76.6	78.0	77.1	74.4	69.7	62.7	57.5	
		MCDB	61.6	64.3	70.2	74.7	81.3	87.2	89.0	87.8	85.6	77.2	68.2	62.5	
5%		WB	52.7	55.2	59.4	64.5	70.5	75.1	77.0	75.7	73.0	67.4	59.7	54.1	
		MCDB	57.2	61.0	67.1	71.9	79.1	85.1	87.8	86.6	83.3	73.8	65.5	58.5	
10%		WB	47.9	50.9	56.3	62.5	68.9	73.8	75.9	74.6	71.4	64.8	56.9	50.0	
		MCDB	53.8	58.1	63.8	69.8	77.0	83.4	86.0	84.9	79.9	72.1	62.4	54.3	
Mean Daily Temperature Range		5% DB	MDBR	18.1	18.6	19.7	20.1	18.7	18.0	18.2	18.8	19.3	19.9	18.7	16.8
			MCDBR	23.1	23.7	24.5	22.6	20.3	20.1	20.8	22.0	21.6	22.2	21.4	21.8
	MCWBR		15.7	16.0	14.7	12.3	9.9	8.0	6.8	7.4	7.9	11.0	14.0	14.9	
	5% WB	MCDBR	19.7	20.5	20.3	18.7	18.4	18.5	18.5	18.6	19.2	18.3	19.3	18.9	
		MCWBR	15.8	16.4	14.5	12.6	10.0	8.5	7.8	7.5	8.0	10.7	15.2	15.2	
Clear-Sky Solar Irradiance	taub	0.301	0.304	0.338	0.376	0.417	0.420	0.442	0.442	0.384	0.342	0.311	0.290		
	taud	2.487	2.486	2.425	2.329	2.264	2.281	2.273	2.248	2.381	2.484	2.523	2.536		
	Ebn at Noon	285	298	295	287	276	274	267	264	276	280	279	283		
	Edh at Noon	26	29	34	40	43	42	43	43	36	29	25	23		
All-Sky Solar Radiation	RadAvg	778	948	1276	1612	1786	2042	2016	1831	1564	1163	836	662		
	RadStd	68	132	75	92	150	175	123	126	121	148	92	83		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page